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Wasif Ullah

Researcher in Evolutionary Computation for Manufacturing Scheduling Optimization

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BRIEF PROFILE

Wasif Ullah is currently a full-time PhD student at the University of Glasgow, James Watt School of Engineering. He has received master's degree in manufacturing systems from university Malaysia Pahang in 2025 and bachelor's degree in mechanical engineering from university of engineering and technology Peshawar, Pakistan in 2022. During his academic career, he worked as a trainee in various multinational production companies and was appointed as a teaching assistant during his master's program. His research interests include Artificial intelligence, Dynamic scheduling, Evolutionary computation and computational optimization particularly in manufacturing systems.

ACADEMIC QUALIFICATION

PhD. (Mechanical Engineering), University of Glasgow, Scotland, UK (Dec 2025 - Ongoing).

MS. Eng. (Manufacturing Systems), Universiti Malaysia Pahang Al-Sultan Abdullah, Malaysia (March 2023-2525).
Master Thesis: Teaching-Learning Based Optimization for Cost-Oriented Hybrid Flow Shop Scheduling.

BS. Eng. Mechanical Engineering, University of Engineering and Technology Peshawar, Pakistan (2018- 22).

RESEARCH INTEREST

1. Artificial Intelligence (AI)
2. Industrial Engineering
3. Modeling and Simulation
4. Metaheuristics
5. Computational Optimization
6. Manufacturing Systems
7. Dynamic Scheduling
8. Multi/Many Objectives Optimization
9. Evolutionary Computation
10. Computational Intelligence

RESEARCH EXPERIENCE

PhD Researcher – University of Glasgow (Dec 2025 – Present)

1. Research on self-learning/composite dispatching rules for dynamic manufacturing scheduling.
2. Focus on evolutionary computation, optimization, AI-driven scheduling, and dynamic manufacturing systems.
3. Development of scheduling models and computational optimization approaches.

Graduate Researcher – Universiti Malaysia Pahang (2023–2025)

1. Research on cost-oriented hybrid flow shop scheduling.
2. Developed metaheuristic optimization approaches including TLBO-based methods.
3. MSc thesis awarded Best Thesis Award.

PUBLICATIONS

Journal Papers

1. **Wasif Ullah**, Mohd Fadzil Faisae Ab Rashid, Muhammad Ammar Nik Mu'tasim, Greedy-assisted teaching-learning-based optimization algorithm for cost-based hybrid flow shop scheduling, *Expert Systems with Applications*, Vol. 273, 2025, 126955. https://doi.org/10.1016/j.eswa.2025.126955 (IF = 7.5)
2. **Ullah, W.**, Mu'tasim, M. A. N., & Rashid, M. F. F. (2024). Optimization of Cost-Based Hybrid Flow Shop Scheduling Using Teaching-Learning-Based Optimization Algorithm. *International Journal of Automotive and Mechanical Engineering*, 21(3), 11616–11628. https://doi.org/10.15282/ijame.21.3.2024.13.0896 (IF = 1.0)

3. **Ullah, W.**, Rashid, M. F. F. Ab., & Mu'tasim, M. A. N., & Tejani, G. G. (2025). Minimizing Total Production Cost in Hybrid Flow Shop Scheduling using Taguchi Enhanced Particle Swarm Optimization Algorithm. *Journal of Advanced Research Design*, 132(1), 41–51. https://doi.org/10.37934/ard.132.1.4151
4. Mohd Fadzil Faisae Ab Rashid, Wasif Ullah, "Optimization of Balanced Academic Curriculum Problem in Educational Institutions Using Teaching Learning Based Optimization Algorithm", IJMECS,17(3),1-14,2025. https://doi.org/10.5815/ijmecs.2025.03.01
5. **Ullah, W.**, Rashid, M. F. F. Ab., & Mu'tasim, M. A. N. (2023). Modeling and Optimization of Cost-Based Hybrid Flow Shop Scheduling Problem using Metaheuristics. *International Journal of Global Optimization and Its Application*, 2(4), 244–254. https://doi.org/10.56225/ijgoia.v2i4.265
6. **Wasif Ullah**, Mohd Fadzil Faisae Ab Rashid and Muhammad Yahya Hammad. Multi-objective dynamic job shop scheduling optimization in manufacturing systems: A short review. International Journal of Science and Research Archive, 2025, 17(02), 885–894. https://doi.org/10.30574/ijsra.2025.17.2.3123

Conference Papers

1. Ab. Rashid, Mohd. F. F., Ullah, W., & Mutasim, M. A. N. (2024). Assessment of Integrated Assembly Sequence Planning and Line Balancing Optimization Using Metaheuristic Algorithms. *2024 IEEE 6th Symposium on Computers & Informatics (ISCI)*, 1, 55–59. https://doi.org/10.1109/isci62787.2024.10668162

Submitted / Under Review / Accepted

1. **Ullah, W.**, Rashid, M. F. F. Ab., & Mu'tasim (Submitted). An Artemisinin Metaheuristic Approach to Minimize Makespan in Hybrid Flow Shop Scheduling with Non-Identical Machines. *Arabian Journal for Science and Engineering* (IF = 2.6).
2. **Ullah, W.**, Rashid, M. F. F. Ab., & Mu'tasim (Submitted). Total Production Cost in Hybrid Flow Shop Scheduling using Differential Evolution (DE) Algorithm. *Journal of Information Systems Engineering and Business Intelligence*.
3. **Ullah, W.**, Rashid, M. F. F. Ab., & Ying Liu (Submitted). Tutor-Guided Optimization for Large-Scale Hybrid Flow Shop Scheduling: A Learning-Inspired Metaheuristic. *Swarm and Evolutionary Computation* (IF = 8.6).
4. Mohd Fadzil Faisae Ab Rashid, Wasif Ullah, Muhammad Ammar Nik Mu'tasim, (Submitted), K-Means Clustering Artemisinin Optimization Algorithm for Non-Identical Hybrid Flow Shop Scheduling, *European Journal of Operational Research* (IF = 6.0)
5. **Ullah, W.**, Rashid, M. F. F. Ab., & Mu'tasim (Accepted), Review of Cost-Based Hybrid Flow Shop Scheduling Optimization: Current Status and Future Directions. *Pakistan Journal of Engineering and Applied Sciences*.
6. Hammad, Y., Syed Radzi Rahamaddulla., Ullah, Wasif. (Submitted). Enhancing SME Awareness of Green Technical Training through Intelligent Manufacturing: A Bibliometric Framework. *Journal of International Education in Business.*
7. Muhammad Ammar Nik Mu'tasim, MFF Ab Rashid, Wasif U, Ying Liu. (Accepted). An Adaptive Elite-Guided Multi-Objective Tiki-Taka Algorithm for Resource-Constrained Project Scheduling with Worker Assignment. *Evolutionary Intelligence*
8. **Wasif Ullah**, Peifeng Li (Submitted). From Dispatching Rules to Self-Learning Scheduling: A Critical Review of Evolutionary and Intelligent Approaches for Dynamic Manufacturing Systems. *Swarm and evolutionary computation*
9. **Wasif Ullah**, Peifeng Li (Submitted). Artificial Intelligence for Dynamic Manufacturing Scheduling: A Systematic Review of Problem Characteristics, Solution Approaches, and Adaptability. *Engineering applications of artificial intelligence*

RESEARCH AND COMPUTATIONAL SKILLS

1. MATLAB
2. Python
3. Optimization and Metaheuristics
4. Modeling and Simulation

TEACHING EXPERIENCES

1. Teaching Assistant – Universiti Malaysia Pahang Al-Sultan Abdullah, Malaysia (03-2023 to 03-2024)

CONFERENCES AND SEMINARS

1. 2024 6th IEEE Symposium on Computers & Informatics (ISCI2024)
2. International Conference of Mechanical Engineering Research (ICMER 2023)

INTERNSHIPS/ INDUSTRIAL EXPERIENCE

1. NSTC IIPP Visiting Research Fellow [NPUST Taiwan] (01/12/2024 – 28/02/2025)
2. Production Trainee [AMF Pakistan Aeronautical Complex KAMRA] (07/Sep/2021-17/Sep/2021)
3. CNC Programmer – GSS Company, Peshawar, Pakistan (06-2019 to 06-2020)

SCIENTIFIC WRITING & VISUALIZATION

1. MS Word
2. Mendeley
3. Papers Formatting
4. LaTeX
5. Proofreading
6. Origin LAB
7. Excell
8. PPT
9. Canva
10. Bio-Render
11. Draw.io
12. Minitab

CAD CAM CAE FEA SKILLS

1. AutoCAD
2. SolidWorks
3. Fusion 360
4. Creo Parametric
5. On Shape
6. Siemens NX
7. Catia

COMPUTER NUMARICAL PROGRAMMING (CNC)

1. CNC Programming (3,4 and 5 Axis) -NX-CAM (Generating G&M codes for part geometry)